

ECE249:BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

L:3 T:0 P:0 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: Understand the workings of various semiconductor devices and analyze their applications.

CO2 :: Familiar to configure Arduino board and sensor module for making application oriented project.

CO3 :: Examine various number systems and their application in digital circuits.

CO4 :: Analyze combinational circuits used in Application-Specific Integrated Circuits.

CO5 :: Evaluate sequential circuits using flip-flops and integrated circuits for practical applications.

CO6 :: Create application-based projects with a focus on skills development and ergonomic design.

Unit I

Fundamentals of Electrical Laws, Semiconductor Devices and its Applications : Ohm's Law, Kirchhoff's Law, Voltage division rule, Current division rule, Basics of semiconductors (Intrinsic and Extrinsic), PN junction diode (working and characteristics) and its applications (rectifiers and switch), Bipolar junction transistor (types, modes, construction, and working CE configuration)

Unit II

Introduction of Arduino and Sensors : Analog and digital signals, Arduino board (pin configuration and description), IR sensor., LDR, basic principle of ultrasonic sensor, Temperature sensor (DHT11/DHT22)

Unit III

Introduction to number system and logic gates : Number system (conversion), codes (B-G, G-B, Excess-3, BCD), Compliments, Binary Arithmetic (addition and subtraction using 2's complement), logic gates, boolean algebra, SOP and POS, K- Map (up to 4 variables)

Unit IV

Introduction to Combinational Logic Circuits : Combinational Logic Circuits: Adders, Subtractors, Multiplexers, De-multiplexers, Decoders, Encoders, Comparator upto 2 bit

Unit V

Introduction to Sequential Logic Circuits : Latch (SR and D), Flip-Flop (SR, JK, D and T), Master-Slave flip-flop, Conversion of basic flip-flop

Unit VI

Applications of Sequential Circuits : Registers: Operation of basic shift registers (SISO, SIPO, PISO, PIPO), Counters: Asynchronous counter (UP/DOWN/Mod-N), Synchronous counter (UP/DOWN/Mod-N), Ring counter and Johnson ring counter

Text Books:

1. DIGITAL LOGIC & COMPUTER DESIGN by M. MORRIS MANO, PEARSON

References:

1. FUNDAMENTALS OF ELECTRICAL ENGINEERING AND ELECTRONICS by B.L.THERAJA, S. CHAND & COMPANY
2. . DIGITAL FUNDAMENTALS BY THOMAS L. FLOYD , R. P JAIN, PEARSON by THOMAS L. FLOYD , R. P JAIN, PEARSON
3. ELECTRONIC CIRCUIT FUNDAMENTALS AND APPLICATIONS by MIKE TOOLEY, NEWNES PUBLISHERS

